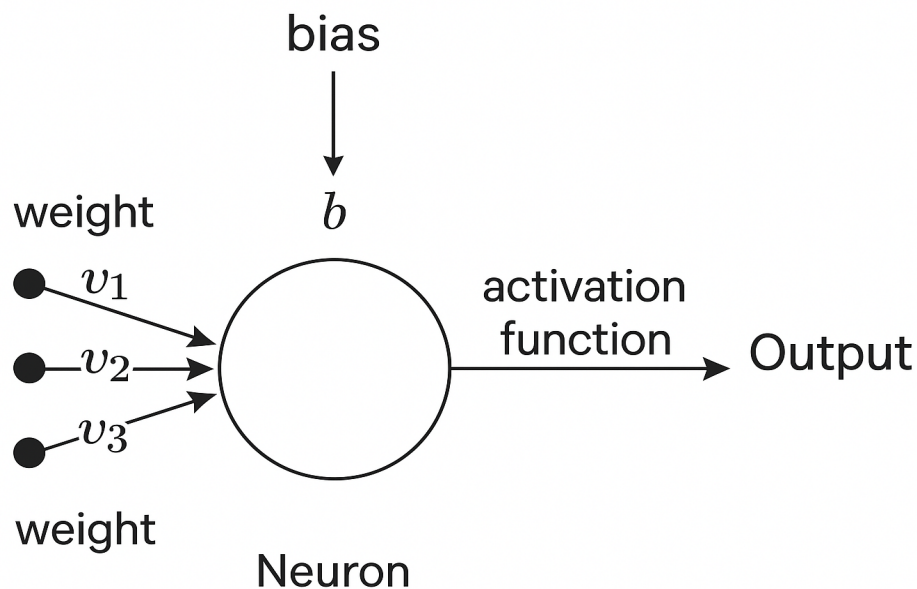


Understanding Neural Networks: Weights, Bias, and Activation Functions

A neural network is made up of small processing units called neurons. Each neuron takes inputs, performs a weighted sum, adds a bias, and passes the result through an activation function to produce an output. 1. **Weight**: Determines how important each input is. It multiplies each input to scale its impact. Example: In predicting rain, cloudiness might have a higher weight than temperature. 2. **Bias**: Allows shifting the output of a neuron up or down. It helps the model make predictions even when all inputs are zero. 3. **Activation Function**: Decides how the neuron reacts. It introduces non-linearity so the model can handle complex data like images or speech. Common activation functions: - ReLU: Outputs $\max(0, z)$ - Sigmoid: Squashes values between 0 and 1 - Tanh: Squashes values between -1 and 1 Together, the neuron performs: $\text{Output} = \text{Activation}(w_1 \cdot x_1 + w_2 \cdot x_2 + \dots + b)$



This diagram shows a single neuron where inputs are multiplied by weights, a bias is added, and the result is passed through an activation function to produce an output.